

PLC & INDUSTRIAL AUTOMATION ENGINEERING PORTFOLIO

Project 5 - Industrial Bottle Filling and Capping System

Version 2 - Industrial Controls Upgrade

PLC Platform: Allen-Bradley MicroLogix 1100

Programming Software: RSLogix Micro Starter Lite

Prepared By: Sharmin Rahman

1. Project Overview

Project 5 Version 2 develops the original conveyor-based production project into a sequential bottle-processing system controlled by an Allen-Bradley MicroLogix 1100 PLC. The final program combines machine start/stop control, operating permissives, bottle detection, conveyor sequencing, timed filling-stage control, timed capping, production counting, and machine-status indication.

I structured the program so the Start push button does not directly control the conveyor or processing outputs. A valid Start command first establishes the latched Machine_Run_Request B3:1/2. This request, together with Start_Permissive B3:0/7 and a selected Auto or Manual mode, establishes System_Ready B3:0/3.

The automatic production sequence uses Conveyor_Request B3:0/2 as an internal command rather than driving Conveyor_Motor O:2/0 directly from the bottle sensor. When the system is ready, Auto mode is selected, the bottle photoeye is clear, and neither filling nor capping is active, the conveyor request is latched.

When Bottle_Photoeye I:1/7 detects a bottle, Filling_Active B3:0/6 is latched and the conveyor request is removed. Fill_Delay T4:1 then controls the filling-stage interval. The final ladder uses a 0.01-second time base and preset of 300, giving a programmed three-second interval.

After the filling delay reaches Done, Filling Active is unlatched and Capping_Active B3:1/0 is latched. Cap_Timer T4:2 then provides a two-second capping interval using a 0.01-second time base and preset of 200. At the end of the capping interval, Capping Active is unlatched and Conveyor Request is relatched so the completed bottle can leave the station and the next cycle can begin. Bottle_Counter C5:0 counts the completed capping event.

One implementation detail is preserved exactly from the final ladder: O:2/1 is labeled Cooling_Fan and is energized during the Filling Active interval before Fill_Delay is done. Earlier support documents call the same output Fill_Valve, but the final ladder screenshot is the governing technical reference.

This project was developed and tested as a PLC training/portfolio implementation rather than commissioned on a physical production machine.

2. Engineering Objective

- Detect invalid simultaneous Auto and Manual mode selection.
- Monitor E-stop and conveyor overload healthy feedback.
- Latch a master machine fault when an implemented abnormal condition occurs.
- Require deliberate fault reset after monitored conditions return healthy.
- Establish a centralized Start Permissive.
- Store the operator machine run request separately from physical outputs.
- Establish a System Ready status from the run request, permissive state, and operating-mode selection.
- Separate conveyor demand from the physical Conveyor Motor output.
- Detect bottle arrival using a photoeye.
- Stop conveyor movement when a bottle reaches the processing position.
- Maintain a filling-stage state independently from the bottle sensor.
- Apply a repeatable three-second filling-stage interval.
- Transfer the process explicitly from filling to capping.
- Apply a repeatable two-second capping interval.
- Restart conveyor movement after capping is complete.
- Count completed capping cycles as completed production.

- Provide running, fault, and audible alarm indication.
- Allow the accumulated production counter to be reset by the operator.

3. Functional Requirements

- Mode_Conflict B3:0/5 shall energize when Auto_Mode I:1/3 and Manual_Mode I:1/4 are selected simultaneously.
- Fault_Active B3:0/4 shall latch for Mode Conflict, EStop_OK loss, or Overload_OK loss.
- Fault Active shall clear only when Reset_PB is true, EStop_OK and Overload_OK are healthy, and Mode Conflict is false.
- Start_Permissive B3:0/7 shall be true only when the implemented machine-health conditions are satisfied.
- Start_PB I:1/0 with Start Permissive shall latch Machine_Run_Request B3:1/2.
- Stop_PB I:1/1 or Fault Active shall unlatch Machine Run Request.
- System_Ready B3:0/3 shall require Start Permissive, Machine Run Request, and Auto or Manual mode.
- In Auto mode, System Ready with bottle photoeye clear and neither process step active shall latch Conveyor_Request B3:0/2.
- Conveyor Request, Bottle Photoeye, and System Ready shall latch Filling_Active B3:0/6.
- Bottle Photoeye shall unlatch Conveyor Request.
- Filling Active with System Ready shall start Fill_Delay T4:1 with time base 0.01 s and preset 300.
- Cooling_Fan O:2/1 shall energize while Filling Active and System Ready are true and T4:1/DN is false.
- T4:1/DN shall unlatch Filling Active and latch Capping_Active B3:1/0 through separate final rungs.
- Capping Active shall start Cap_Timer T4:2 with time base 0.01 s and preset 200.
- Cap_Solenoid O:2/2 shall energize while Capping Active is true and T4:2/DN is false.
- T4:2/DN shall unlatch Capping Active and relatch Conveyor Request.
- Conveyor_Motor O:2/0 shall require Conveyor Request, System Ready, and Fault Active false.
- Bottle_Counter C5:0 shall count the Cap Timer Done event with preset 9999.
- Running_Lamp O:2/3 shall follow System Ready.
- Fault_Lamp O:2/4 and Alarm_Horn O:2/5 shall follow Fault Active.
- Reset_PB I:1/2 shall execute RES C5:0.

4. System Components

Component	Identification	Function
PLC Platform	Allen-Bradley MicroLogix 1100	Executes machine sequence and fault logic
Programming Software	RSLogix Micro Starter Lite	PLC programming and simulation/testing
Start Push Button	I:1/0	Latches machine run request
Stop Push Button	I:1/1	Unlatches machine run request
Reset Push Button	I:1/2	Fault and counter reset command
Auto Mode Selector	I:1/3	Enables automatic conveyor sequence
Manual Mode Selector	I:1/4	Participates in operating authorization
EStop_OK	I:1/5	Healthy feedback monitored by fault/permissive logic
Overload_OK	I:1/6	Conveyor overload healthy feedback
Bottle Photoeye	I:1/7	Bottle detection at station
Conveyor Motor	O:2/0	Conveyor command
Cooling Fan	O:2/1	Final screenshot label for output active during filling interval

Cap Solenoid	O:2/2	Capping actuator command
Running Lamp	O:2/3	System Ready indication
Fault Lamp	O:2/4	Master fault indication
Alarm Horn	O:2/5	Audible master fault indication
Fill Delay	T4:1	Three-second filling-stage timer
Cap Timer	T4:2	Two-second capping timer
Bottle Counter	C5:0	Counts completed cap cycles

5. I/O Assignment

Physical Inputs

Device	PLC Address	Type	Description
Start_PB	I:1/0	Digital Input	Machine start command
Stop_PB	I:1/1	Digital Input	Machine stop command
Reset_PB	I:1/2	Digital Input	Fault/counter reset command
Auto_Mode	I:1/3	Digital Input	Automatic mode selection
Manual_Mode	I:1/4	Digital Input	Manual mode selection
EStop_OK	I:1/5	Digital Input	E-stop healthy feedback
Overload_OK	I:1/6	Digital Input	Conveyor overload healthy feedback
Bottle_Photoeye	I:1/7	Digital Input	Bottle detection at station

Physical Outputs

Device	PLC Address	Type	Description
Conveyor_Motor	O:2/0	Digital Output	Conveyor motor command
Cooling_Fan	O:2/1	Digital Output	Final screenshot label for filling-interval output
Cap_Solenoid	O:2/2	Digital Output	Capping actuator command
Running_Lamp	O:2/3	Digital Output	System Ready indication
Fault_Lamp	O:2/4	Digital Output	Fault indication
Alarm_Horn	O:2/5	Digital Output	Audible fault indication

Internal PLC Data

Function	Address	Type	Purpose
Conveyor_Request	B3:0/2	Latched internal bit	Requests conveyor movement
System_Ready	B3:0/3	Internal bit	Machine authorization status
Fault_Active	B3:0/4	Latched internal bit	Master fault state
Mode_Conflict	B3:0/5	Internal bit	Simultaneous Auto/Manual selection
Filling_Active	B3:0/6	Latched internal bit	Filling process state
Start_Permissive	B3:0/7	Internal bit	Combined start permission
Capping_Active	B3:1/0	Latched internal bit	Capping process state
Machine_Run_Request	B3:1/2	Latched internal bit	Stored operator run request
Fill_Delay	T4:1	TON	0.01 s base; PRE 300
Cap_Timer	T4:2	TON	0.01 s base; PRE 200
Bottle_Counter	C5:0	CTU	Completed cap-cycle counter; PRE 9999

The earlier written ladder document describes Counter_OneShot B3:1/1, but no ONS instruction appears in the final PV5 counter rung. The final implementation therefore does not document B3:1/1 as an active counter element.

6. Control Strategy

I organized Project 5 Version 2 around Machine Health -> Run Authorization -> Conveyor Request -> Filling State -> Capping State -> Completed Production. This separates machine authorization, sequence decisions, physical outputs, and production information.

Mode Conflict, E-stop feedback loss, and overload feedback loss can latch Fault Active. Start Permissive then provides a single machine-health status. Start and Start Permissive latch Machine Run Request, while Stop or Fault Active removes it. System Ready combines that maintained request with Start Permissive and an Auto or Manual mode selection.

The automatic conveyor sequence is controlled through Conveyor Request rather than directly writing the motor. When Auto mode is selected, the station is clear, and neither processing state is active, Conveyor Request latches. Bottle detection then latches Filling Active and unlatches Conveyor Request so the bottle remains stationary.

Fill_Delay T4:1 times the three-second filling-stage interval. In the final ladder, Cooling_Fan O:2/1 is energized while Filling Active and System Ready are true and the Fill Delay has not completed. T4:1/DN then terminates Filling Active and starts Capping Active using separate state-transition rungs.

Cap_Timer T4:2 provides the two-second capping interval. Cap Solenoid is energized while Capping Active is true and T4:2/DN is false. When the timer completes, Capping Active is unlatched and Conveyor Request is relatched. The physical conveyor output also rechecks System Ready and Fault Active at the output level.

Bottle Counter is triggered by T4:2/DN. The final screenshot does not use a separate one-shot; the CTU counts the false-to-true transition of its rung condition.

7. Engineering Design Decisions

Design Decision	Engineering Reason
Dedicated Mode Conflict	Makes invalid selector operation explicit
Latched Fault Active	Requires deliberate recovery after faults
Central Start Permissive	Provides one machine-health permission point
Latched Machine Run Request	Allows momentary Start to establish sustained authorization
System Ready separate from Conveyor Motor	Keeps machine status active during intentional process pauses
Internal Conveyor Request	Separates sequence demand from physical motor output
Filling Active state bit	Maintains filling sequence after bottle detection
Bottle Photoeye unlatches Conveyor Request	Stops the bottle at the station
Independent Fill Delay	Provides repeatable three-second interval
Separate fill-end and cap-start rungs	Makes step transition explicit
Independent Capping Active state	Maintains cap operation through timer interval
Independent Cap Timer	Provides repeatable two-second capping interval
Cap Solenoid with XIO T4:2/DN	Limits actuation to the capping interval
Conveyor Request relatched after capping	Automatically releases completed bottle and continues cycle
Fault Active in Conveyor Motor rung	Provides final output-level fault inhibition
Bottle Counter from cap completion	Counts fully processed cycles rather than detected bottles
Running Lamp from System Ready	Remains on during filling/capping pauses

Two implementation details are intentionally preserved from the final ladder: O:2/1 is labeled Cooling_Fan rather than Fill_Valve, and the final Bottle Counter rung does not contain the previously documented ONS B3:1/1.

8. Ladder Logic Description

8.1 Rung 0000 - Mode Conflict Detection

```
|----[ XIC I:1/3 ]----[ XIC I:1/4 ]----- ( OTE B3:0/5 )----|
      Auto_Mode      Manual_Mode      Mode_Conflict
```

Rung Description: When Auto and Manual are both true, Mode Conflict energizes. Because OTE is used, the bit follows the current selector condition.

Engineering Purpose: I included this rung to explicitly identify invalid simultaneous mode selection.

8.2 Rung 0001 - Master Fault Latch

```
+----[ XIC B3:0/5 ]-----+
|      Mode_Conflict      |
+----[ XIO I:1/5 ]-----+----( OTL B3:0/4 )----|
|      EStop_OK          |      Fault_Active
+----[ XIO I:1/6 ]-----+
      Overload_OK
```

Rung Description: Mode Conflict, loss of EStop_OK, or loss of Overload_OK can latch the master Fault Active bit.

Engineering Purpose: The latched master fault prevents automatic recovery after a transient abnormal condition.

8.3 Rung 0002 - Fault Reset

```
|----[ XIC I:1/2 ]----[ XIC I:1/5 ]----[ XIC I:1/6 ]----[ XIO B3:0/5 ]----( OTU B3:0/4 )----|
      Reset_PB      EStop_OK      Overload_OK      Mode_Conflict      Fault_Active
```

Rung Description: Reset unlatches Fault Active only while E-stop and overload feedback are healthy and no Mode Conflict exists.

Engineering Purpose: Conditional reset prevents the operator from clearing the master fault while a monitored abnormal condition remains active.

8.4 Rung 0003 - Start Permissive

```
|----[ XIO B3:0/4 ]----[ XIC I:1/5 ]----[ XIC I:1/6 ]----[ XIO B3:0/5 ]----- ( OTE B3:0/7 )--
--|
      Fault_Active      EStop_OK      Overload_OK      Mode_Conflict
Start_Permissive
```

Rung Description: Start Permissive is true only while the implemented health conditions are satisfied.

Engineering Purpose: This creates one centralized permission bit for machine startup.

8.5 Rung 0004 - Machine Run Request Latch

```
|----[ XIC I:1/0 ]----[ XIC B3:0/7 ]----- ( OTL B3:1/2 )----|
      Start_PB      Start_Permissive      Machine_Run_Request
```

Rung Description: A valid Start command latches Machine Run Request.

Engineering Purpose: The internal latch separates the operator request from physical outputs and keeps the machine authorized after the momentary Start is released.

8.6 Rung 0005 - Machine Run Request Unlatch

```

+-----[ XIC I:1/1 ]-----+
|-----+      Stop_PB      +----- ( OTU B3:1/2 )-----|
+-----[ XIC B3:0/4 ]-----+      Machine_Run_Request
      Fault_Active

```

Rung Description: Stop or Fault Active removes the stored machine run request.

Engineering Purpose: One centralized unlatch rung provides predictable normal-stop and fault-stop behavior.

8.7 Rung 0006 - System Ready

```

|-----[ XIC B3:0/7 ]-----[ XIC B3:1/2 ]-----+-----[ XIC I:1/3 ]-----+----- ( OTE B3:0/3 )-----|
      Start_Permissive  Machine_Run_Request  |      Auto_Mode      |      System_Ready
+-----[ XIC I:1/4 ]-----+
      Manual_Mode

```

Rung Description: System Ready requires the permissive, the stored machine request, and either Auto or Manual mode.

Engineering Purpose: This status keeps machine authorization distinct from individual sequence states and physical outputs.

8.8 Rung 0007 - Automatic Conveyor Request

```

|-----[ XIC B3:0/3 ]-----[ XIC I:1/3 ]-----[ XIO I:1/7 ]-----[ XIO B3:0/6 ]-----[ XIO B3:1/0 ]----- ( OTL B3:0/2 )-----|
      System_Ready      Auto_Mode      Bottle_Photoeye      Filling_Active      Capping_Active      Conveyor_Request

```

Rung Description: In Auto mode, the conveyor request latches when the station is clear and neither processing state is active.

Engineering Purpose: The internal request separates sequence logic from the physical conveyor output.

8.9 Rung 0008 - Filling Sequence Start

```

|-----[ XIC B3:0/2 ]-----[ XIC I:1/7 ]-----[ XIC B3:0/3 ]----- ( OTL B3:0/6 )-----|
      Conveyor_Request  Bottle_Photoeye      System_Ready      Filling_Activ

```

Rung Description: Bottle detection while the conveyor is requested and the system is ready latches Filling Active.

Engineering Purpose: The latched state lets the filling-stage sequence continue after the sensor event.

8.10 Rung 0009 - Conveyor Request Removal

```

|-----[ XIC I:1/7 ]----- ( OTU B3:0/2 )-----|
      Bottle_Photoeye      Conveyor_Request

```

Rung Description: When the bottle photoeye becomes true, Conveyor Request is unlatched.

Engineering Purpose: This stops bottle movement at the processing position.

8.11 Rung 0010 - Fill Delay

```

|-----[ XIC B3:0/6 ]-----[ XIC B3:0/3 ]-----[ TON T4:1 ]-----|
      Filling_Active      System_Ready      Fill_Delay  Time Base: 0.01 Preset: 300

```

Rung Description: Fill Delay accumulates while Filling Active and System Ready are true. The programmed interval is three seconds.

Engineering Purpose: The timer provides a repeatable filling-stage duration.

8.12 Rung 0011 - Filling-Interval Output

```
|----[ XIC B3:0/6 ]----[ XIC B3:0/3 ]----[ XIO T4:1/DN ]----- ( OTE O:2/1 )----|
      Filling_Active      System_Ready      Fill_Delay_Done      Cooling_Fan
```

Rung Description: The final screenshot-labeled Cooling Fan output is energized during the active filling interval before Fill Delay reaches Done.

Engineering Purpose: This rung ties the processing output to both the active state and timer window. The screenshot label is preserved exactly.

8.13 Rung 0012 - End Filling State

```
|----[ XIC T4:1/DN ]----- ( OTU B3:0/6 )----|
      Fill_Delay_Done      Filling_Active
```

Rung Description: Fill Delay Done unlatches Filling Active.

Engineering Purpose: The rung explicitly terminates the filling state at the end of its timed interval.

8.14 Rung 0013 - Start Capping State

```
|----[ XIC T4:1/DN ]----- ( OTL B3:1/0 )----|
      Fill_Delay_Done      Capping_Active
```

Rung Description: The same Fill Delay Done event latches Capping Active.

Engineering Purpose: A separate rung makes the transition from filling to capping easy to monitor and troubleshoot.

8.15 Rung 0014 - Cap Timer

```
|----[ XIC B3:1/0 ]-----[ TON T4:2 ]----|
      Capping_Active      Cap_Timer Time Base: 0.01 Preset: 200
```

Rung Description: Cap Timer accumulates while Capping Active is true. The programmed interval is two seconds.

Engineering Purpose: An independent timer makes capping duration separate from filling duration.

8.16 Rung 0015 - Cap Solenoid Output

```
|----[ XIC B3:1/0 ]----[ XIO T4:2/DN ]----- ( OTE O:2/2 )----|
      Capping_Active      Cap_Timer_Done      Cap_Solenoid
```

Rung Description: Cap Solenoid is energized while Capping Active is true and Cap Timer is not done.

Engineering Purpose: The timer-not-done condition limits actuator energization to the active capping interval.

8.17 Rung 0016 - End Capping State

```
|----[ XIC T4:2/DN ]----- ( OTU B3:1/0 )----|
      Cap_Timer_Done      Capping_Active
```

Rung Description: Cap Timer Done unlatches Capping Active.

Engineering Purpose: This explicitly ends the capping state at the completion of the timed interval.

8.18 Rung 0017 - Restart Conveyor Request

```
|----[ XIC T4:2/DN ]----- ( OTL B3:0/2 )----|
      Cap_Timer_Done      Conveyor_Request
```

Rung Description: Cap Timer Done also relatches Conveyor Request.

Engineering Purpose: The completed bottle can leave the station and the next automatic cycle can begin.

8.19 Rung 0018 - Conveyor Motor Output

```
|----[ XIC B3:0/2 ]----[ XIC B3:0/3 ]----[ XIO B3:0/4 ]----- ( OTE O:2/0 )----|
      Conveyor_Request   System_Ready      Fault_Active           Conveyor_Motor
```

Rung Description: The physical conveyor output requires Conveyor Request and System Ready and is inhibited by Fault Active.

Engineering Purpose: This keeps the physical motor command behind both sequence demand and final machine authorization.

8.20 Rung 0019 - Completed Bottle Counter

```
|----[ XIC T4:2/DN ]-----[ CTU C5:0 ]----|
      Cap_Timer_Done           Bottle_Counter PRE: 9999
```

Rung Description: The CTU increments from the Cap Timer Done event. The final screenshot does not show an ONS instruction.

Engineering Purpose: Counting at cap completion represents completed production rather than bottle arrival or partial processing.

8.21 Rung 0020 - Running Lamp

```
|----[ XIC B3:0/3 ]----- ( OTE O:2/3 )----|
      System_Ready           Running_Lamp
```

Rung Description: Running Lamp follows System Ready.

Engineering Purpose: Using machine authorization instead of the conveyor output keeps the running indication active while the bottle is stationary for processing.

8.22 Rung 0021 - Fault Lamp

```
|----[ XIC B3:0/4 ]----- ( OTE O:2/4 )----|
      Fault_Active           Fault_Lamp
```

Rung Description: Fault Lamp follows the latched master fault state.

Engineering Purpose: This provides persistent visual fault indication.

8.23 Rung 0022 - Alarm Horn

```
|----[ XIC B3:0/4 ]----- ( OTE O:2/5 )----|
      Fault_Active           Alarm_Horn
```

Rung Description: Alarm Horn follows Fault Active.

Engineering Purpose: The audible indication attracts attention when a fault occurs.

8.24 Rung 0023 - Bottle Counter Reset

```
|----[ XIC I:1/2 ]-----[ RES C5:0 ]----|
      Reset_PB           Bottle_Counter
```

Rung Description: Reset PB clears the Bottle Counter accumulator and status bits.

Engineering Purpose: This supports production-count reset for testing or a new counting period.

8.25 Rung 0024 - END

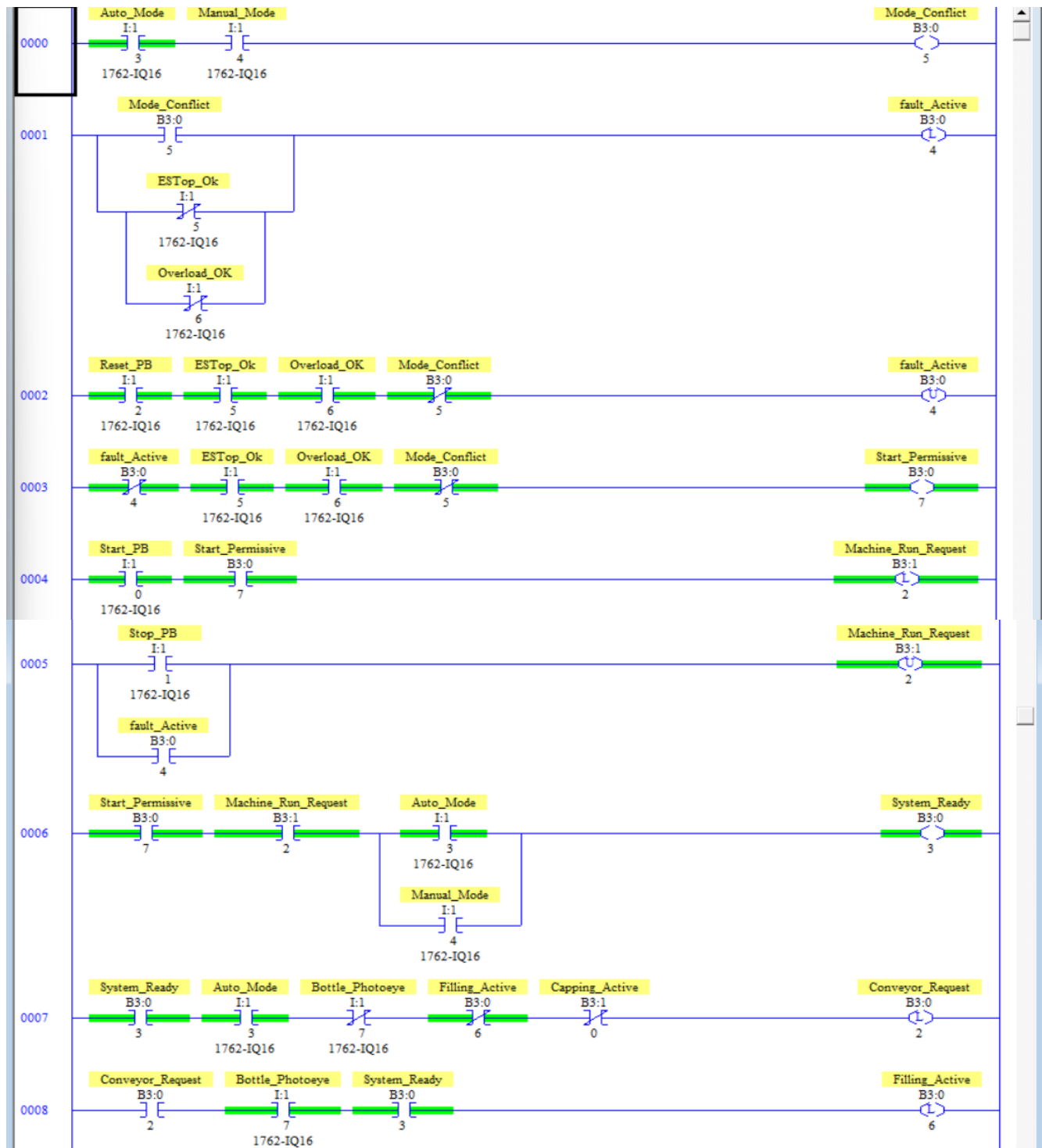
```
|----- ( END )-----|
```

Rung Description: END marks the end of the ladder program.

Engineering Purpose: This closes the main program file.

8.26 Actual RSLogix Implementation

The following images are the final PV5 ladder screenshots and are the absolute technical source of truth for the project. If any typed representation appears ambiguous, the screenshot implementation governs.



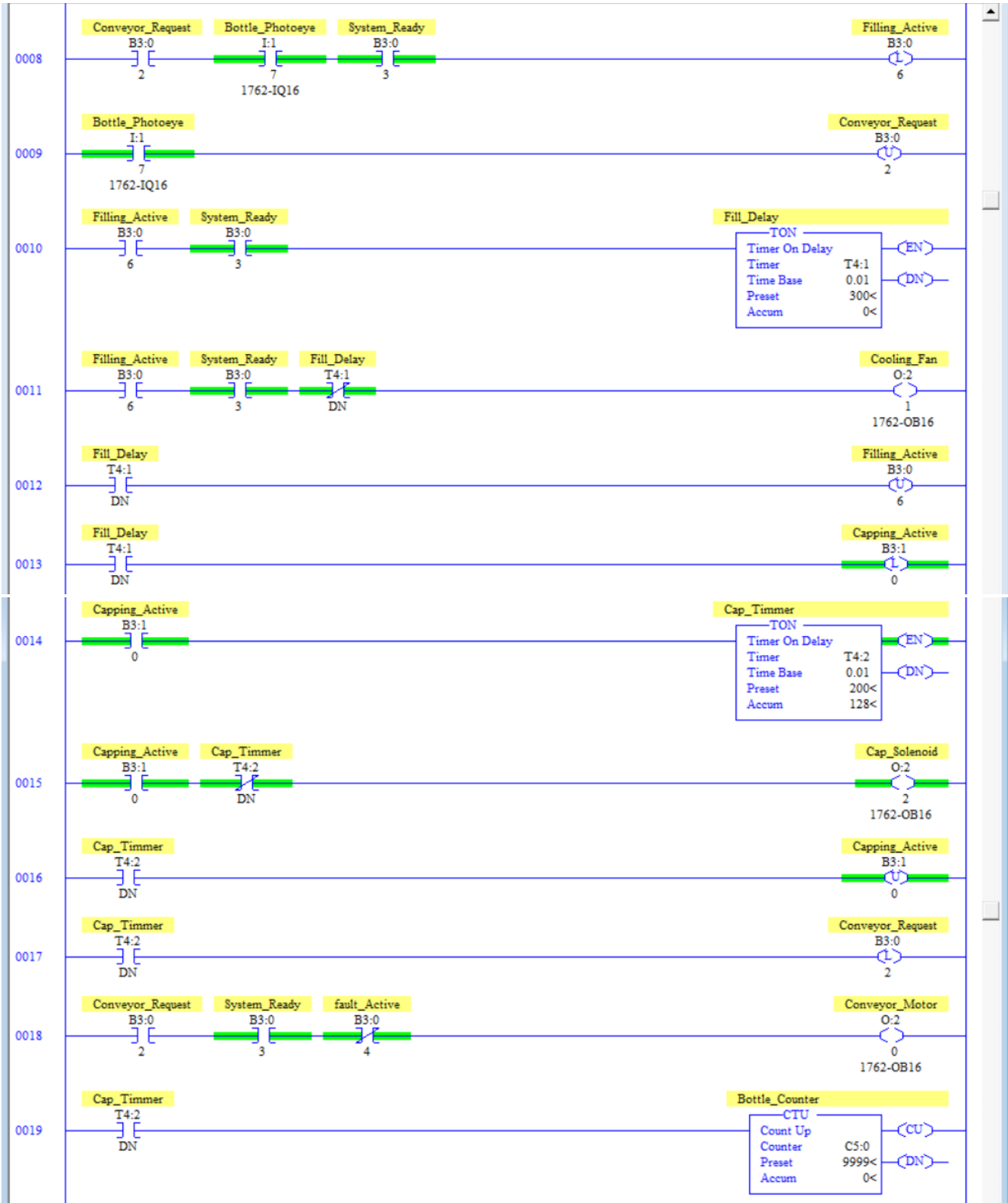




Figure P5V2.5 - Final RSLogix ladder implementation, part 5

9. Sequence of Operation

Step 1 - Initial Condition: The PLC evaluates Mode Conflict, E-stop healthy feedback, overload healthy feedback, and Fault Active before machine authorization.

Step 2 - Start Permission: Healthy conditions establish Start_Permissive B3:0/7.

Step 3 - Machine Start: Pressing Start latches Machine_Run_Request B3:1/2.

Step 4 - System Ready: Start Permissive and Machine Run Request with Auto or Manual selected establish System_Ready B3:0/3.

Step 5 - Automatic Conveyor Request: In Auto mode, with the station clear and neither process state active, Conveyor_Request B3:0/2 latches.

Step 6 - Conveyor Runs: Conveyor Request, System Ready, and no Fault Active energize Conveyor_Motor O:2/0.

Step 7 - Bottle Detection: Bottle_Photoeye I:1/7 becomes true at the station.

Step 8 - Filling State: Conveyor Request, Bottle Photoeye, and System Ready latch Filling_Active B3:0/6.

Step 9 - Conveyor Stops: Bottle Photoeye unlatches Conveyor Request, removing the motor command.

Step 10 - Fill Timing: Filling Active and System Ready enable T4:1 for the programmed three-second interval.

Step 11 - Filling-Interval Output: Cooling_Fan O:2/1 is energized while Filling Active and System Ready remain true and T4:1/DN is false.

Step 12 - Transfer to Capping: T4:1/DN unlatches Filling Active and latches Capping Active.

Step 13 - Cap Timing: Capping Active enables T4:2 for the programmed two-second interval.

Step 14 - Cap Actuation: Cap Solenoid O:2/2 remains energized until T4:2/DN becomes true.

Step 15 - End Capping: T4:2/DN unlatches Capping Active.

Step 16 - Restart Conveyor: T4:2/DN relatches Conveyor Request.

Step 17 - Production Count: The same T4:2/DN event increments C5:0 as implemented in the final rung.

Step 18 - Continuous Cycle: With System Ready maintained, the conveyor moves the completed bottle away and the sequence repeats for the next bottle.

Step 19 - Stop / Fault: Stop or Fault Active removes Machine Run Request and System Ready, which removes controlled outputs.

Step 20 - Counter Reset: Reset_PB executes RES C5:0 when production count reset is required.

10. Testing & Validation

Test ID	Test Action	Expected Result	Observed Result	Status
T-01	Healthy conditions + valid mode	Start_Permissive becomes true	Permissive established	PASS
T-02	Press Start	Machine_Run_Request latches	Run request latched	PASS
T-03	Auto mode with station clear	Conveyor_Request and Conveyor_Motor energize	Conveyor sequence started	PASS
T-04	Activate Bottle_Photoeye	Filling_Active latches; Conveyor_Request unlatches	Bottle stopped and filling state entered	PASS
T-05	Observe T4:1	Timer completes 3.00 s programmed interval	Fill timing completed	PASS
T-06	Observe O:2/1 during fill interval	Screenshot-labeled Cooling_Fan is active before T4:1/DN	Output followed final ladder	PASS
T-07	Allow T4:1/DN	Filling_Active unlatches; Capping_Active latches	State transfer completed	PASS
T-08	Observe T4:2	Timer completes 2.00 s programmed interval	Cap timing completed	PASS
T-09	Observe O:2/2	Cap_Solenoid active while Capping_Active and timer not done	Cap output followed sequence	PASS
T-10	Allow T4:2/DN	Capping_Active unlatches and Conveyor_Request relatches	Conveyor restart command established	PASS
T-11	Complete cap cycle	C5:0 increments from final counter rung	Completed bottle count increased	PASS
T-12	Press Stop during operation	Machine_Run_Request drops and controlled outputs stop	Stop behavior verified	PASS
T-13	Make EStop_OK false	Fault_Active latches; machine operation stops	E-stop monitoring response verified	PASS
T-14	Make Overload_OK false	Fault_Active latches; machine operation stops	Overload response verified	PASS
T-15	Select Auto and Manual together	Mode_Conflict and Fault_Active activate	Mode conflict response verified	PASS
T-16	Attempt/perform fault reset	Reset succeeds only after healthy conditions are restored	Controlled reset verified	PASS
T-17	Press Reset for production counter	C5:0 resets	Counter reset verified	PASS

All defined requirements were verified on the final tested PLC implementation. Testing represents PLC program validation in the development/simulation environment, not physical packaging-machine commissioning or safety-system validation.

11. Results

Project 5 Version 2 successfully implemented a repeatable sequence that authorizes the machine, transports a bottle to the station, stops the conveyor, performs a timed filling-stage interval, transfers to a timed capping interval, counts completed production, and restarts the conveyor for the next cycle.

The control architecture separates machine authorization from physical outputs through Machine Run Request, System Ready, Conveyor Request, Filling Active, and Capping Active. Fault handling is centralized through Fault Active, while final Conveyor Motor logic also checks that the fault state is clear.

The final implementation passed the defined functional tests in the PLC development/simulation environment. The report preserves the final screenshot implementation, including the O:2/1 Cooling_Fan label and the CTU counter rung without a separate ONS instruction.

12. Challenges & Troubleshooting

A useful diagnostic order is: Fault / Permissive -> Machine Run Request -> System Ready -> Conveyor Request -> Bottle Photoeye -> Filling Active -> Fill Delay -> Capping Active -> Cap Timer -> Conveyor Motor / Counter.

Symptom	Possible Cause	Diagnostic Check	Corrective Action
Machine will not start	Fault active, unhealthy E-stop/overload, or mode conflict	Check B3:0/4, B3:0/5, I:1/5, I:1/6, B3:0/7	Restore healthy conditions and perform valid reset
Run request will not latch	Start input or Start Permissive missing	Monitor I:1/0 and B3:0/7	Verify Start and permissive chain
Conveyor will not run	System Ready or Conveyor Request false; Fault Active true	Check B3:0/3, B3:0/2, B3:0/4, O:2/0	Trace authorization and sequence request
Conveyor does not stop at bottle	Photoeye not changing or request not unlatching	Monitor I:1/7, B3:0/6, B3:0/2	Verify photoeye and rungs 0008-0009
Filling interval does not start	Filling Active/System Ready missing	Check B3:0/6, B3:0/3, T4:1/EN	Trace bottle-detection state transition
O:2/1 does not energize	Filling Active/System Ready false or T4:1/DN already true	Check B3:0/6, B3:0/3, T4:1/DN, O:2/1	Verify final screenshot rung 0011
Capping does not start	T4:1/DN not true or Capping Active not latching	Check T4:1/DN and B3:1/0	Trace rungs 0012-0013
Cap solenoid does not energize	Capping Active false or T4:2/DN already true	Check B3:1/0, T4:2/DN, O:2/2	Verify capping state and timer
Sequence does not restart after capping	Conveyor Request not relatched or System Ready dropped	Check T4:2/DN, B3:0/2, B3:0/3	Trace rungs 0017-0018
Bottle counter does not increment	T4:2/DN event not occurring or counter reset held	Check T4:2/DN, C5:0.ACC, I:1/2	Verify cap completion and reset state
Fault lamp/horn remain on	Fault Active remains latched	Check B3:0/4 and fault sources	Restore healthy conditions and perform valid reset

The troubleshooting guide supplied with the project provides useful diagnostic sequencing, but any reference to a Fill_Valve at O:2/1 or to a counter one-shot must be interpreted against the final PV5 ladder because the final screenshot implementation differs on those points.

13. Lessons Learned

This project strengthened my understanding of state-based sequential control. Instead of using one bottle sensor to directly switch several outputs, I used internal state bits to represent the machine run request, conveyor request, filling stage, and capping stage.

I learned how separate timers can control independent process steps and how explicit state transitions make a sequence easier to monitor online. The project also reinforced the value of keeping physical outputs behind internal requests and final authorization conditions.

Counting at the completion of the capping interval showed how production data can be tied to a meaningful process milestone rather than to an earlier sensor event. I also learned the importance of comparing final PLC screenshots against earlier written documentation before publishing a portfolio report.

14. Industrial Relevance

Bottle filling and capping sequences are common in packaging, food and beverage, pharmaceutical, chemical, and consumer-product production. The project demonstrates relevant concepts such as machine permissives, latched run requests, sensor-based indexing, timed process steps, state transitions, conveyor interlocks, fault handling, status indication, and production counting.

The project remains a PLC training/simulation implementation. A real production cell could additionally require dedicated bottle-position and exit sensors, product-level or flow feedback, cap-present verification, guarding, safety-rated hardware, conveyor feedback, reject handling, HMI functions, recipe management, and commissioning against the actual process.

15. Version 1 -> Version 2 Engineering Progression

Version 1 Foundation	Version 2 Implementation	Engineering Benefit
Basic conveyor control	Machine Run Request + System Ready	Separates operator demand from equipment command
Simple bottle detection	Latched Filling Active state	Maintains process state after sensor event
Direct conveyor behavior	Internal Conveyor Request	Separates sequence demand from physical motor output
Single/simple processing step	Explicit Filling and Capping states	Creates a traceable multi-step sequence
Basic timing	Independent T4:1 and T4:2 timers	Separates fill and cap durations
Limited interlocks	Fault, permissive, mode, and output checks	Improves shutdown and troubleshooting structure
Limited production information	Completed Bottle Counter C5:0	Tracks completed cycles
Simple status indication	Running, Fault, and Alarm outputs	Improves operator visibility
Foundation logic	Explicit state transition rungs	Improves maintainability and online diagnostics

The main progression is not simply more rungs. Version 2 separates authorization, conveyor movement, filling, capping, and completed production into identifiable control states that can be monitored and troubleshoot independently.

16. Project Summary

Project 5 Version 2 developed the conveyor foundation into an Industrial Bottle Filling and Capping System using the Allen-Bradley MicroLogix 1100 platform. The final program includes mode-conflict detection, master fault handling, Start Permissive, Machine Run Request, System Ready, Conveyor Request, bottle detection, Filling Active, three-second Fill Delay, Capping Active, two-second Cap Timer, Cap Solenoid control, conveyor restart, completed-bottle counting, and machine-status outputs.

The sequence demonstrates my progression from basic conveyor logic toward a structured PLC machine sequence using internal request bits, explicit process states, timers, output interlocks, fault handling, and production information. All defined requirements were tested successfully in the PLC development/simulation environment.

The final documentation follows the PV5 screenshots as the absolute technical source of truth. In particular, it preserves the O:2/1 Cooling_Fan label and the Bottle Counter rung without the previously documented one-shot instruction.